

AI Assisted Coding

Assignment-6.3

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Batch-02

Task:01

Task Description #1 (Loops – Automorphic Numbers in a Range)

- Task: Prompt AI to generate a function that displays all Automorphic numbers between 1 and 1000 using a for loop.

- Instructions:

- o Get AI-generated code to list Automorphic numbers using a for loop.

- o Analyze the correctness and efficiency of the generated logic.

- o Ask AI to regenerate using a while loop and compare both implementations.

Expected Output #1:

- Correct implementation that lists Automorphic numbers using both loop types, with explanation.

```
automorphic.py > ...  
1  #display all automorphic numbers within 1 to 1000  
2  for num in range(1, 1001):  
3      square = num ** 2  
4      if str(square).endswith(str(num)):  
5          print(num)
```

Outputs:

```
PS C:\Users\nithi\Desktop\ai assisted coding> & C:/Users/nithi/AppData/Local/Programs/Python/Python312/python.exe "c:/Users/nithi/Desktop/ai assisted coding/automorphic.py"
1
5
6
25
76
376
625
PS C:\Users\nithi\Desktop\ai assisted coding>
```

Using while loop:

```
automorphic.py > ...
2  num = 1
3  while num <= 1000:
4      square = num ** 2
5      if str(square).endswith(str(num)):
6          print(num)
7      num += 1
```

Output:

```
PS C:\Users\nithi\Desktop\ai assisted coding> & C:/Users/nithi/AppData/Local/Programs/Python/Python312/python.exe "c:/Users/nithi/Desktop/ai assisted coding/automorphic.py"
1
5
6
25
76
376
625
PS C:\Users\nithi\Desktop\ai assisted coding>
```

The difference between while and for loop:

Best Choice for Automorphic Numbers 1-1000: FOR LOOP

Why:

1. You **know exact bounds** (1 to 1000) upfront
2. **Fewer lines** and no manual increment needed
3. **Pythonic** - aligns with Python's design philosophy
4. **No infinite loop risk** - safer code

```
on automorphic_timing_comparison.py
1 5 6 25 76 376 625

1 5 6 25 76 376 625

=====
TIMING COMPARISON
=====
For Loop Time: 0.00000000 seconds
While Loop Time: 0.00103951 seconds
Difference: 0.00103951 seconds
Winner: For Loop
PS C:\Users\nithi\Desktop\ai assisted coding>
```

Task Description #2 (Conditional Statements – Online Shopping Feedback Classification)

- Task: Ask AI to write nested if-elif-else conditions to classify online shopping feedback as Positive, Neutral, or Negative based on a numerical rating (1–5).

- Instructions:

- o Generate initial code using nested if-elif-else.
- o Analyze correctness and readability.
- o Ask AI to rewrite using dictionary-based or match-case structure.

Expected Output #2:

- Feedback classification function with explanation and an alternative approach.

```

online_shopping.py > ...
1  #write a python nested if-elif-else conditions to classify online
2  #shopping feedback Positive, Neutral, or Negative based on a numeric rating (1-5).
3  #using dictionary to map ratings to feedback
4
5  rating = int(input("Enter your rating (1-5): "))
6
7  feedback_map = {
8      1: "Negative",
9      2: "Negative",
10     3: "Neutral",
11     4: "Positive",
12     5: "Positive"
13 }
14
15 if rating in feedback_map:
16     print(f"Feedback: {feedback_map[rating]}")
17 else:
18     print("Invalid rating. Please enter a number between 1 and 5.")

```

Outputs:

```

PS C:\Users\nithi\Desktop\ai assisted coding> & C:/Users/nithi/AppData/Local/Programs/Python/Python312/python.exe "c:/Users/nithi/Desktop/ai assisted coding/online_shopping.py"
Enter your rating (1-5): 3
Feedback: Neutral
PS C:\Users\nithi\Desktop\ai assisted coding> & C:/Users/nithi/AppData/Local/Programs/Python/Python312/python.exe "c:/Users/nithi/Desktop/ai assisted coding/online_shopping.py"
Enter your rating (1-5): 4
Feedback: Positive
PS C:\Users\nithi\Desktop\ai assisted coding>

```

Task 3: Statistical_operations

Define a function named `statistical_operations(tuple_num)` that performs the following statistical operations on a tuple of numbers:

- Minimum, Maximum
- Mean, Median, Mode
- Variance, Standard Deviation

While writing the function, observe the code suggestions provided by GitHub Copilot. Make decisions to accept, reject, or modify the suggestions based on their relevance and correctness

```

statistical_operations.py > ...
1  # Define a function named statistical_operations(tuple_num) that performs the
2  # following statistical operations on a tuple of numbers:
3  # • Minimum, Maximum
4  # • Mean, Median, Mode
5  # • Variance, Standard Deviation
6  import statistics
7  def statistical_operations(tuple_num):
8      if not tuple_num:
9          return None
10
11     minimum = min(tuple_num)
12     maximum = max(tuple_num)
13     mean = statistics.mean(tuple_num)
14     median = statistics.median(tuple_num)
15     try:
16         mode = statistics.mode(tuple_num)
17     except statistics.StatisticsError:
18         mode = "No unique mode"
19     variance = statistics.variance(tuple_num)
20     std_dev = statistics.stdev(tuple_num)
21
22     return {
23         "Minimum": minimum,
24         "Maximum": maximum,
25         "Mean": mean,
26         "Median": median,
27         "Mode": mode,
28         "Variance": variance,
29         "Standard Deviation": std_dev
30     }
31
32 numbers = (1, 2, 2, 3, 4)
33 result = statistical_operations(numbers)
34 print(result)

```

Output:

```

312/python.exe "c:/Users/nithi/Desktop/ai assisted coding/statistical_operations.py"
{'Minimum': 1, 'Maximum': 4, 'Mean': 2.4, 'Median': 2, 'Mode': 2, 'Variance': 1.3, 'Standard Deviation': 1.140175425099138}
PS C:\Users\nithi\Desktop\ai assisted coding> 

```

Task 4: Teacher Profile

- Prompt: Create a class Teacher with attributes teacher_id, name, subject, and experience. Add a method to display teacher details.
- Expected Output: Class with initializer, method, and object creation.

```

teacher_profile.py > ...
1  # • Prompt: Create a class Teacher with attributes teacher_id, name,
2  # subject, and experience. Add a method to display teacher details.
3  # • Expected Output: Class with initializer, method, and object creation.
4  class Teacher:
5      def __init__(self, teacher_id, name, subject, experience):
6          self.teacher_id = teacher_id
7          self.name = name
8          self.subject = subject
9          self.experience = experience
10
11     def display_details(self):
12         print(f"Teacher ID: {self.teacher_id}")
13         print(f"Name: {self.name}")
14         print(f"Subject: {self.subject}")
15         print(f"Experience: {self.experience} years")
16 # Creating an object of the Teacher class
17 teacher1 = Teacher(101, "Nithin", "Mathematics", 10)
18 teacher1.display_details()
19 teacher2 = Teacher(102, "Abhilash", "Science", 8)
20 teacher2.display_details()
21

```

Output:

```

PS C:\Users\nithi\Desktop\ai assisted coding> & C:/Users/nithi/AppData/Local/Programs/Python/Python
312/python.exe "c:/Users/nithi/Desktop/ai assisted coding/teacher_profile.py"
Teacher ID: 101
Name: Nithin
Subject: Mathematics
Experience: 10 years
Teacher ID: 102
Name: Abhilash
Subject: Science
Experience: 8 years
PS C:\Users\nithi\Desktop\ai assisted coding> 

```

Task #5 – Zero-Shot Prompting with Conditional Validation

Use zero-shot prompting to instruct an AI tool to generate a function that validates an Indian mobile number.

Requirements

- The function must ensure the mobile number:
 - o Starts with 6, 7, 8, or 9
 - o Contains exactly 10 digits

Expected Output

- A valid Python function that performs all required validations without using any input-output examples in the prompt.

```
indian_number.py > ...
1  #generate a function that validates an Indian phone number
2  #The function should check if the number starts with 7, 8, or 9 and is exactly 10 digits long.
3  def validate_indian_phone_number(phone_number):
4      if len(phone_number) == 10 and phone_number.isdigit() and phone_number[0] in ['7', '8', '9']:
5          return True
6      else:
7          return False
8
9  phone_number = input("Enter an Indian phone number: ")
10 if validate_indian_phone_number(phone_number):
11     print("Valid Indian phone number.")
12 else:
13     print("Invalid Indian phone number.")
```

Output:

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
> & C:/Users/nithi/AppData/Local/Programs/Python/Python312/python.exe "c:/Users/nithi/Desktop/ai
Enter an Indian phone number: 9704035106
Valid Indian phone number.
PS C:\Users\nithi\Desktop\ai assisted coding> |
```